

- 3 The photograph shows a marine worm, *Nereis pelagic*, extending its body from a burrow in the sand.



(Source: © Frank Hecker/Alamy Stock Photo)

Magnification $\times 1$

This animal is found in the Sea of Japan. This animal feeds on organic particles in mud and sand habitats.

A scientist observed that these animals respond to a sudden increase in light intensity by withdrawing into their burrows.

The animals then extend their bodies to continue feeding.

The scientist selected 11 animals living in mud and 11 animals living in sand.

Each animal was exposed to one burst of bright light.

The time in seconds before beginning to extend their bodies from their burrows was then recorded.

Mud habitat

14.8 8.2 9.9 11.6 12.0 14.2 10.7 11.2 8.9 12.1 12.9

Sand habitat

14.0 7.4 9.2 12.6 10.7 7.8 8.0 11.1 10.7 10.8 11.2

(a) State a suitable null hypothesis for this investigation.

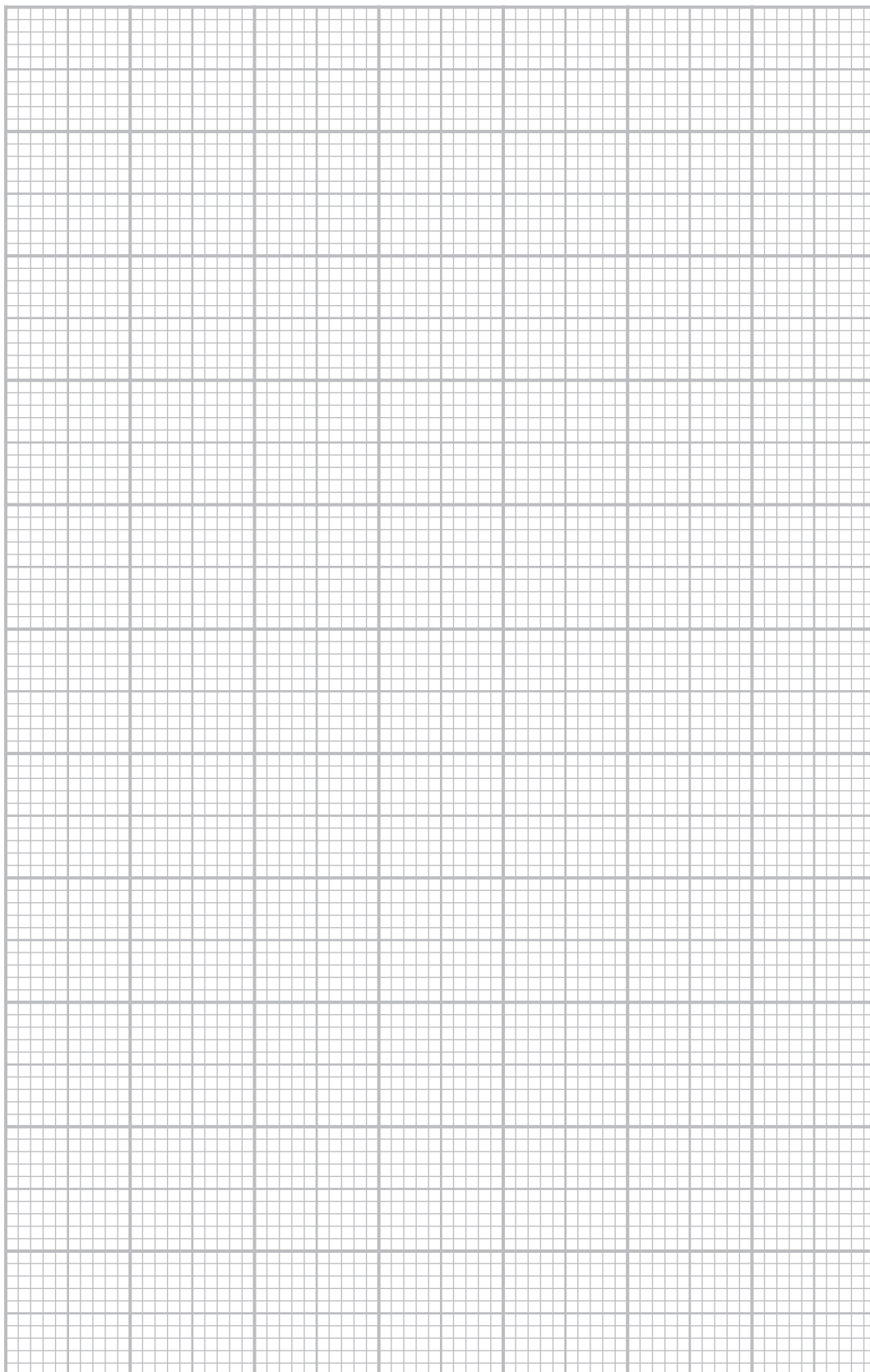
(1)

(b) (i) Draw a suitable table to display the **data** and your calculated **means** for time before extension in mud and sand habitats.

(3)

(b) (ii) Draw a suitable graph to display your calculated **means** and the variability of the data, for these two groups of animals.

(3)



(c) The student analysed the data with a t test using the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$ is the mean value for each habitat

n is the number of samples for each habitat

$(S_A)^2 = 4.17$ and $(S_B)^2 = 4.22$

(i) Calculate the value of t .

(3)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

Number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	p = 0.05	p = 0.01
15	2.13	2.95
16	2.12	2.92
17	2.11	2.90
18	2.10	2.88
19	2.09	2.86
20	2.09	2.84
21	2.08	2.83
22	2.07	2.82
23	2.07	2.81
24	2.06	2.80
25	2.06	2.79
26	2.06	2.78
27	2.05	2.77
28	2.05	2.76
29	2.04	2.76
30	2.04	2.75

Explain the conclusion that can be drawn from this investigation.

Use your graph, your calculated t value and the table of critical values to support your answer.

(3)

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(d) This animal has light sensitive pigments called opsins.

Suggest how opsins allow the animals to respond in this investigation.

(2)

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(Total for Question 3 = 15 marks)